

ENGINEERING REPORT

Custom PCB Power System Architecture

AquaSense Automatic Touchless Water Dispenser
Power Distribution, Regulation, Protection and Power Integrity

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This report documents the proposed custom PCB power architecture for supplying the controller, display, sensor, and dual-servo actuator subsystem. Component ratings, battery protection, thermal performance, and final measurements must be verified before physical deployment.

1. System Overview and Power Requirements

The AquaSense PCB is designed to support an Arduino Uno R3, an I2C 1602 LCD, an HC-SR04 ultrasonic sensor, and two high-torque servos. The power architecture uses two 18650 Li-ion cells in series, providing approximately 6.0–8.4 V, with power entering through screw terminal J5 and passing through slide switch SW1.

Parameter	Specification
Input source	2S 18650 Li-ion battery pack
Input range	6.0–8.4 V
Power entry	Screw terminal J5
Main isolation	Manual slide switch SW1

2. Power Distribution and Regulation

After switching, the battery input feeds two separate buck-converter paths. The logic/sensor rail at J6 regulates the battery input to 5 V for the Arduino Uno, LCD at J2, and HC-SR04 at J1. The actuator rail at J7 supplies the dual servos at J3 and J4, isolating their high-current transients from sensitive logic and sensor circuitry.

Rail	Loads	Purpose
Logic/sensor — J6	Arduino, LCD J2, HC-SR04 J1	Stable 5 V logic and sensing supply.
Actuator — J7	Dual servos J3 and J4	Dedicated high-current servo supply.

3. Filtering and Power Integrity

The PCB uses localized capacitors to reduce ripple and support transient current demand. C3 is a 470 μF electrolytic bulk capacitor positioned near the servo and sensor connection blocks. C1 is a 10 μF electrolytic capacitor for regulator input/output filtering, and C2 is a 100 nF ceramic capacitor for high-frequency bypassing.

The BATTERY_SW+, EXTERNAL_+5V, and SERVO_V+ paths use approximately 0.8–1.0 mm traces to reduce resistive voltage drop and thermal buildup. Resistive loss follows $P = I^2R$, so reducing trace resistance is especially important in battery and servo paths.

4. Grounding and EMC

The two-layer PCB uses a continuous bottom-layer B.Cu GND copper pour. This provides a low-impedance return path, improves reference stability, reduces loop area, and helps limit the effect of electromagnetic interference and switching noise. Component placement and return-current routing must still be reviewed so that high-current servo returns do not unnecessarily disturb sensitive sensor references.

5. PCB Interconnection Summary

Reference	Function
J1	HC-SR04 ultrasonic sensor.
J2	I2C 1602 LCD.
J3, J4	Dual servo motors.
J5	Battery input screw terminal.
J6	Logic/sensor 5 V buck output.
J7	Dedicated servo buck output.
SW1	Manual power isolation.
C1, C2, C3	Filtering and transient suppression capacitors.

6. Protection and Verification

- Verify battery polarity and use an appropriate protected 2S Li-ion arrangement.
- Measure both buck-converter outputs before connecting loads.
- Confirm the logic rail is suitable for the Arduino, LCD, and sensor.
- Confirm the servo rail supports peak current without unacceptable voltage sag.
- Check capacitor polarity, voltage ratings, and ripple-current capability.
- Inspect continuity, shorts, connector orientation, and switch isolation.
- Measure converter and trace temperature during sustained servo operation.
- Review clearances around battery, switching, and high-current nodes.
- Add fuse or resettable protection if required by final load analysis.

7. Design Specification Summary

Subsystem	Specification / value	Purpose
Input	6.0–8.4 V, 2S 18650	Primary energy source.
Switch	SW1 manual slide switch	Hardware-level shutoff.
Logic buck	J6 dedicated 5 V output	Arduino, LCD, and sensor supply.
Servo buck	J7 dedicated output	High-current actuator supply.
Capacitors	470 μ F + 10 μ F + 100 nF	Bulk, ripple, and high-frequency filtering.
PCB	2-layer; front signals, back GND	Low-impedance grounding and thermal support.
Power traces	0.8–1.0 mm approximate width	Reduced drop and heating.

8. Recommended Commissioning Sequence

- Inspect component orientation and PCB workmanship.
- Check continuity and unintended shorts.
- Connect the battery through J5 with SW1 OFF and verify polarity.
- Measure switched battery voltage after SW1.
- Configure and measure the logic buck output.
- Configure and measure the servo buck output.
- Connect logic loads and verify stable operation.
- Test one servo, then both servos, while observing voltage response.
- Record voltage, current, temperature, ripple observations, resets, and sensor errors.

9. Conclusion

The AquaSense custom PCB power system combines a switched 2S Li-ion input, independent buck-regulated logic and servo rails, localized filtering, high-current trace routing, and a continuous ground plane. This architecture is intended to provide stable power to the Arduino Uno, LCD, HC-SR04 sensor, and dual servos while reducing the effect of actuator transients on sensitive electronics. Final qualification requires verification of battery protection, converter settings, peak servo current, capacitor ratings, thermal behavior, clearances, and measured voltage stability.